

Condensate Partition Coefficient Pipeline

Instruction Manual

Automated 3D segmentation of nuclear biomolecular condensates and measurement of the partition coefficient from confocal Z-stacks

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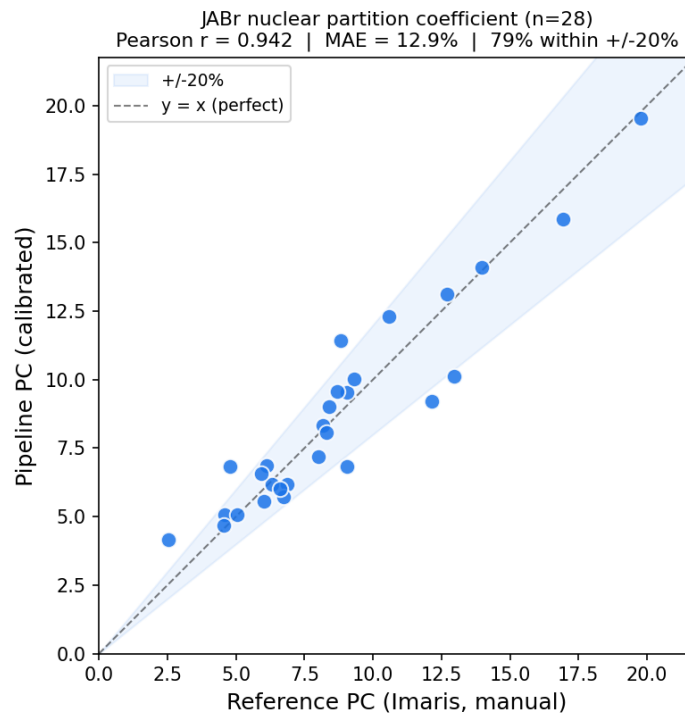
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Overview

This pipeline replaces the manual Imaris workflow for measuring the **partition coefficient (PC)** of nuclear biomolecular condensates. Given a two-channel confocal Z-stack, it segments nuclei and condensates in 3D, computes the PC, and reports a calibrated value on the manual reference scale.

Item	Description
Inputs	Two-channel Z-stack: channel 0 = nuclei, channel 1 = condensate.
Outputs	Nuclear & cytoplasmic PC, 3D masks, per-object volumes, summary figure, interactive zviewer.html .
Validated on	JABr: Pearson $r = 0.942$, MAE 12.9%, 79% of cells within $\pm 20\%$ ($n = 28$).
Interfaces	Web app (webapp.py), desktop GUI (run_gui.py), or command line (pipeline.py).



JABr validation: calibrated pipeline PC vs manual Imaris reference.

Installation (one time)

1. Install Python 3.10 or 3.11.
2. Install PyTorch for your machine from pytorch.org (CPU or your CUDA version), e.g. CUDA 12.1:

```
pip install torch --index-url https://download.pytorch.org/whl/cu121
```

3. Install the remaining dependencies:

```
pip install -r requirements.txt
```

On first run, Cellpose downloads its model weights automatically (one-time ~100 MB). A GPU is recommended (~10–15 s per cell); CPU works but is slower.

Preparing your image files

The pipeline accepts either one multi-channel TIF (both channels in one file) or two separate single-channel TIFs. **Channel order matters:** channel 0 must be nuclei, channel 1 the condensate. Most axis orderings (ZCYX, CZYX, ...) are auto-detected. If results look wrong, swapped channels are the most common cause.

Running with the web app

The web app is the most integrated interface: configure, run, and view results in a single browser page, for both single cells and whole batches. Launch with `python webapp.py`; it opens `http://127.0.0.1:5000` automatically. Pick an input (single TIF, separate channels, single-channel, or a batch folder) using the in-page file browser, set Construct = JABr, and click Run Pipeline. Progress streams live, and the interactive Z-viewer loads in the right-hand panel when the run finishes. In batch mode each cell becomes a clickable chip showing its calibrated PC; clicking it loads that cell's viewer, and `r / RMSE / MAE` appear when a reference CSV is supplied.

Running with the graphical interface (desktop)

A native desktop alternative to the web app. Launch with `python run_gui.py`. A window with **Single File** and **Batch** tabs opens.

Step by step (single cell)

1. On the **Single File** tab, choose the input mode and Browse to your data.
2. Choose an Output folder (optional).
3. In **Settings**, set **Construct = JABr**; leave all other settings at their defaults.
4. Click  **Run Pipeline**. Progress streams in the Output Log; the status reads **Done** when finished.
5. Open your output folder for the masks, tables, and results.png.

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Single File

Batch

- ☒ Single multi-channel TIF (OME / Cut ROI)
- ☐ Separate condensate + nuclei files
- ☐ Single channel — segment one image, nuclei or condensate (no PC)

Multi-Channel TIF

TIF file:

Output

Output folder:

Top-X% brightest voxels for cond. density: (default 75 — recommended)
Nuclei cell probability threshold: (default -2.0)
Construct: (picks calibration + auto-routes detector)
Condensate detector: (auto: JABr→blob_log, else Cellpose)
Cellpose cond. model path:
(blank = built-in cyto3; recommended: V3 epoch 35)
blob_log threshold: (default 0.03 — only used when detector=blob_log)
☐ Disable GPU (use on laptop / no CUDA)
☒ Write interactive Z-viewer (zviewer.html)

▶ Run Pipeline

Output Log



The Single-File tab configured for a JABr run: a multi-channel TIF loaded, Construct = JABr.

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Single File Batch

☒ Single multi-channel TIF (OME / Cut ROI)
☐ Separate condensate + nuclei files
☐ Single channel — segment one image, nuclei or condensate (no PC)

Multi-Channel TIF
TIF file: Browse...

Output
Output folder: Browse...

Top-X% brightest voxels for cond. density: (default 75 — recommended)
Nuclei cell probability threshold: (default -2.0)
Construct: (picks calibration + auto-routes detector)
Condensate detector: (auto: JABr→blob_log, else Cellpose)
Cellpose cond. model path:
(blank = built-in cyto3; recommended: V3 epoch 35)
blob_log threshold: (default 0.03 — only used when detector=blob_log)
☐ Disable GPU (use on laptop / no CUDA)
☒ Write interactive Z-viewer (zviewer.html)

▶ Run Pipeline Done ✓

Output Log

```
Open my_output/zviewer.html in any browser.  
All outputs saved to: my_output
```

After the run: the Output Log reports the calibrated partition coefficient and the status reads Done.

Single-channel mode. A third input mode segments just one image — a condensate *or* a nuclei image — when you do not have both channels. Nuclei use Cellpose cyto3; condensate uses the same detector the full pipeline would (for JABr, blob_log, with the intra-nuclear gate disabled so every detection is kept). It outputs masks, object count, volumes, and measurements — but no partition coefficient, which requires both channels.

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Single FileBatch

Single multi-channel TIF (OME / Cut ROI)

Separate condensate + nuclei files

Single channel — segment one image, nuclei or condensate (no PC)

Single Channel

Image file:

C:\storage\code\condensate-partition-pipeline\sam

Browse...

This image is:condensate

(segments this one channel; outputs masks + measuremen

Output

Output folder:my_output

Browse...

Top-X% brightest voxels for cond. density:

75

(default 75 — recommended)

Nuclei cell probability threshold:

-2.0

(default -2.0)

Construct:

JABr

(picks calibration + auto-routes detector)

Condensate detector:

auto

(auto: JABr→blob_log, else Cellpose)

Cellpose cond. model path:(blank = built-in cyto3; recommended: V3 epoch 35)

blob_log threshold:

0.03

(default 0.03 — only used when detector=blob_log)

☐ Disable GPU (use on laptop / no CUDA)

☒ Write interactive Z-viewer (zviewer.html)

▶ Run Pipeline

Output Log

Single-channel mode: segment one condensate (or nuclei) image on its own.

Understanding the settings

For routine JABr analysis you only need to set **Construct**; the rest have sensible defaults.

Setting	Default	What it does
Construct	(none)	Pick JABr for the validated workflow. Auto-selects the blob_log detector and the JABr calibration.
Condensate detector	auto	auto routes JABr→blob_log, else Cellpose. Normally left on auto.
Top-X% brightest	75	Defines condensed-phase density; trims the dim mask halo to match the manual reference.
Nuclei cell-prob.	-2.0	How aggressively nuclei are detected. -2.0 suits the lab's stain.
blob_log threshold	0.03	Condensate spot sensitivity (used when detector = blob_log). Leave at 0.03 for JABr.

Setting	Default	What it does
Cellpose cond. model	(blank)	Advanced; a fine-tuned condensate model. Leave blank for JABr.
Disable GPU	off	Tick on a laptop with no NVIDIA GPU.
Write interactive Z-viewer	on	Writes a self-contained zviewer.html for scrolling the Z-stack in a browser. Untick to skip.

The one rule for routine use: set Construct = JABr and run. Everything else is for experimentation or other constructs.

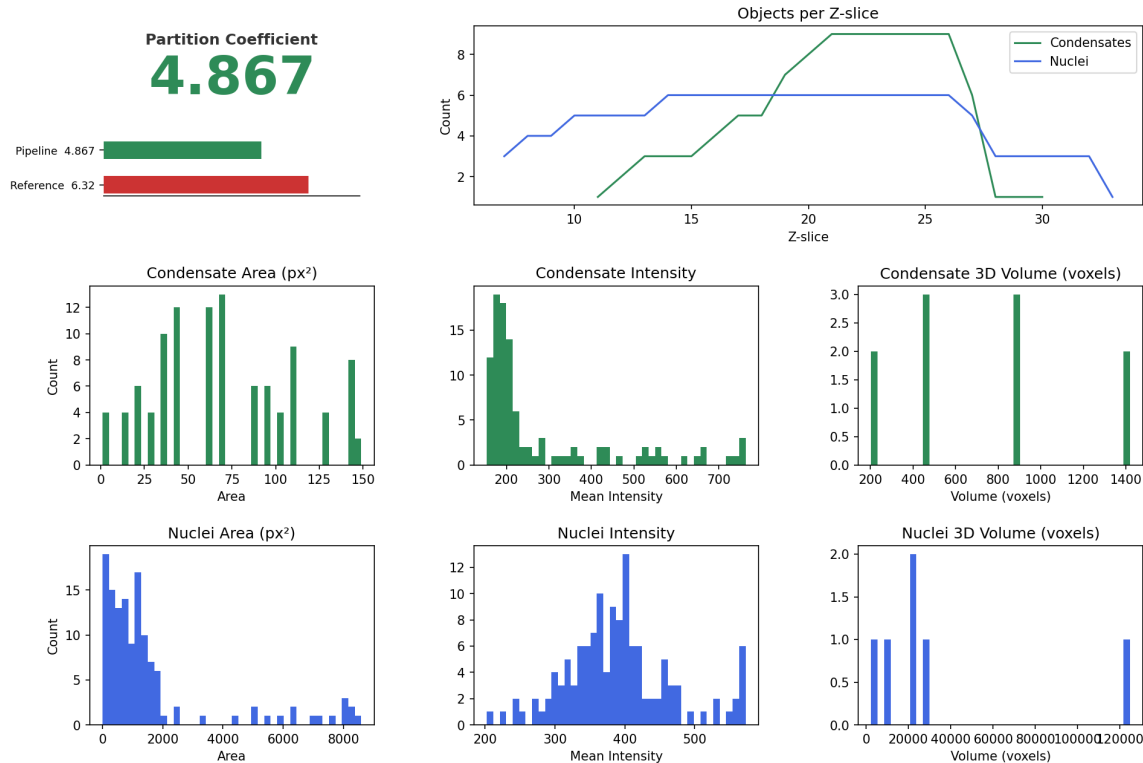
Reading the outputs

Every run writes summary.csv (headline numbers incl. the calibrated PC), results.png (summary figure), condensate_masks.tif and nuclei_masks.tif (3D labeled masks), per-object/per-slice CSVs, and zviewer.html.

The interactive Z-viewer (zviewer.html). Double-click it to open in any browser — it is self-contained (no Python, no internet) and shareable. Drag the slider or use the arrow keys to scroll Z-slices. For each slice it shows four panels (raw; masks-on-row with nuclei = magenta, condensate = yellow outlines; condensate mask; nuclei mask), the partition coefficient and stack summary, live per-slice counts/areas/intensities, and two in-mask intensity histograms (current slice and whole stack). In batch mode each cell gets its own viewer.

For the bundled sample JABr_Sample2_5_3.tif, a real run produces **nuclear PC raw 4.867** → **calibrated 4.813** (manual reference 4.558, +5.6%).

Report the calibrated nuclear PC. The raw value is systematically ~3× the manual scale by construction; calibration standardizes it. **Always glance at condensate_masks.tif** over the raw channel in Fiji/ImageJ — a 10-second check catches the rare misfire.



results.png from the bundled-sample run.

Batch mode and command line

On the **Batch** tab, select a folder of .tif files; every file is processed. Optionally supply the manual Imaris nuclear-PC CSV as a Reference CSV — the pipeline then writes comparison.csv and scatter.png with the correlation, RMSE, and MAE.

Command line:

```
python pipeline.py --roi sample_data/JABr_Sample2_5_3.tif --construct JABr --output my_output
```

Separate channels: --nuc nuclei.tif --cond condensate.tif. Single-channel segmentation (one image, no PC): --single condensate.tif --channel condensate (or --channel nuclei). Physical volumes in μm^3 : add --voxel-xy 0.065 --voxel-z 0.3. Also write the interactive viewer: --view. Force CPU: --no-gpu.

Troubleshooting

Symptom	Likely cause / fix
PC wildly off / nuclei empty	Channels swapped: confirm ch0 = nuclei, ch1 = condensate.
"0 condensates detected"	Not JABr, or very dim signal. Lower blob_log threshold (e.g. 0.02); confirm the raw channel has s
Very slow (minutes/cell)	Running on CPU. Install a CUDA PyTorch build; ensure Disable GPU is unchecked.

Symptom	Likely cause / fix
CUDA out of memory	Tick Disable GPU, or close other GPU programs.
Cellpose download fails	First run needs internet for model weights.
One cell's calibrated PC off	Calibration is population-level; cells vary ($\pm 20\%$ for $\sim 80\%$ of cells). Verify against the mask.

Method summary and scope

The PC is the ratio of condensed-phase to dilute-phase density (Fabrini et al.), both camera-background-subtracted. Condensed density uses the brightest 75% of voxels inside ($\text{condensate} \cap \text{nucleus}$); dilute density is the mean of the 50 lowest-intensity $10 \times 10 \times 10$ patches in the nuclear dilute region. Nuclei are segmented with Cellpose 3 (cyto3, 3D), cleaned, and void-filled; condensates are detected with Laplacian-of-Gaussian blob_log and kept when $\geq 50\%$ of the rendered sphere lies inside a nucleus. A per-construct linear (JABr) or isotonic calibration standardizes the automated PC onto the manual scale.

Scope. Validated and production-ready for JABr. The detector is construct-specific; a leave-one-construct-out test confirmed a single detector does not generalize zero-shot (it under-detects on unseen constructs). Extending to other constructs needs per-construct calibration or few-shot retraining. Full rationale and the cross-construct evidence are in docs/METHODS.md.